

LECTURE 4 — ENERGY GENERATION

PHY199-RAP · Energy in the 21st Century

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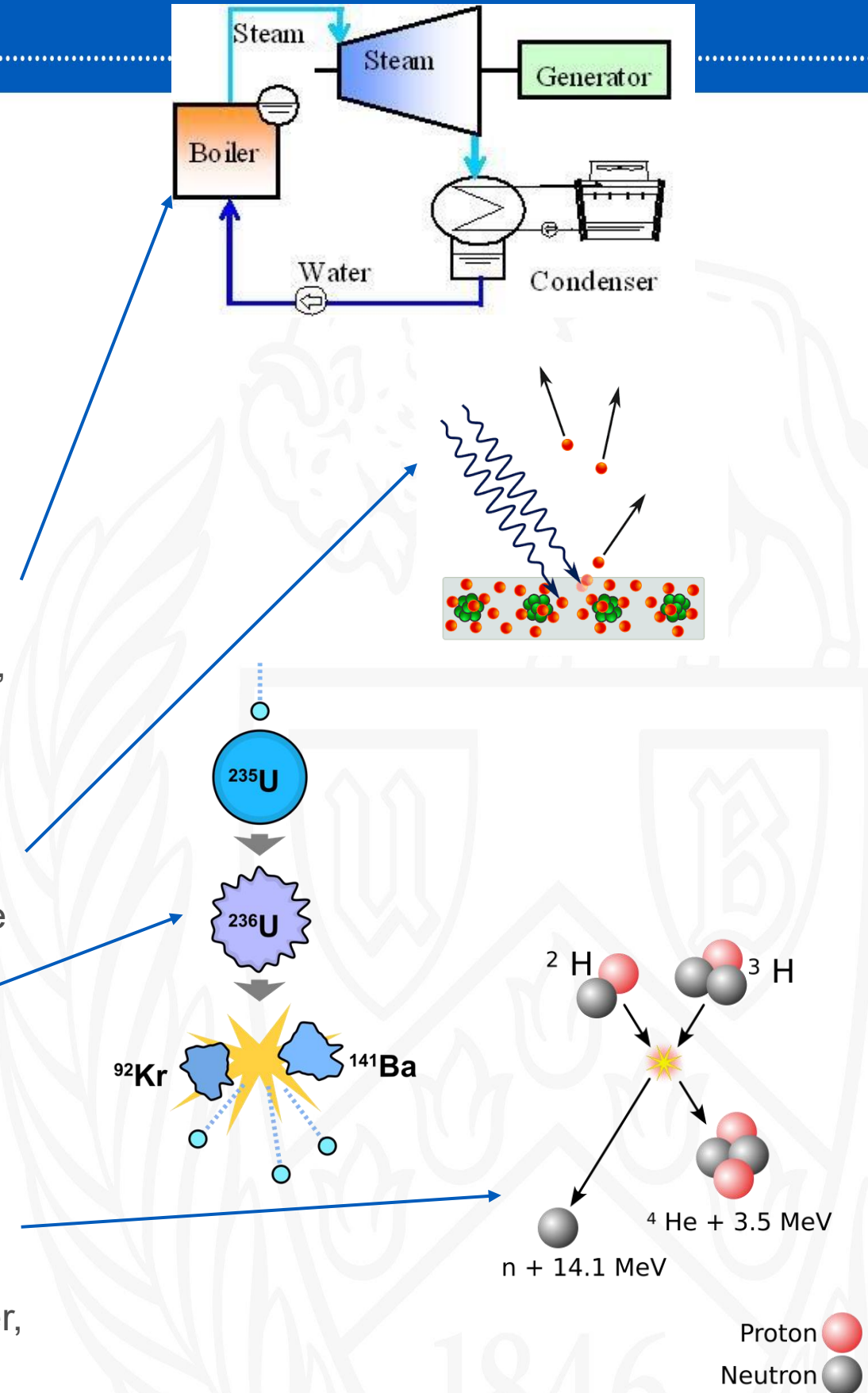
1846

PHYSICS OF POWER GENERATION SOURCES



The Physics of Power Generation Sources

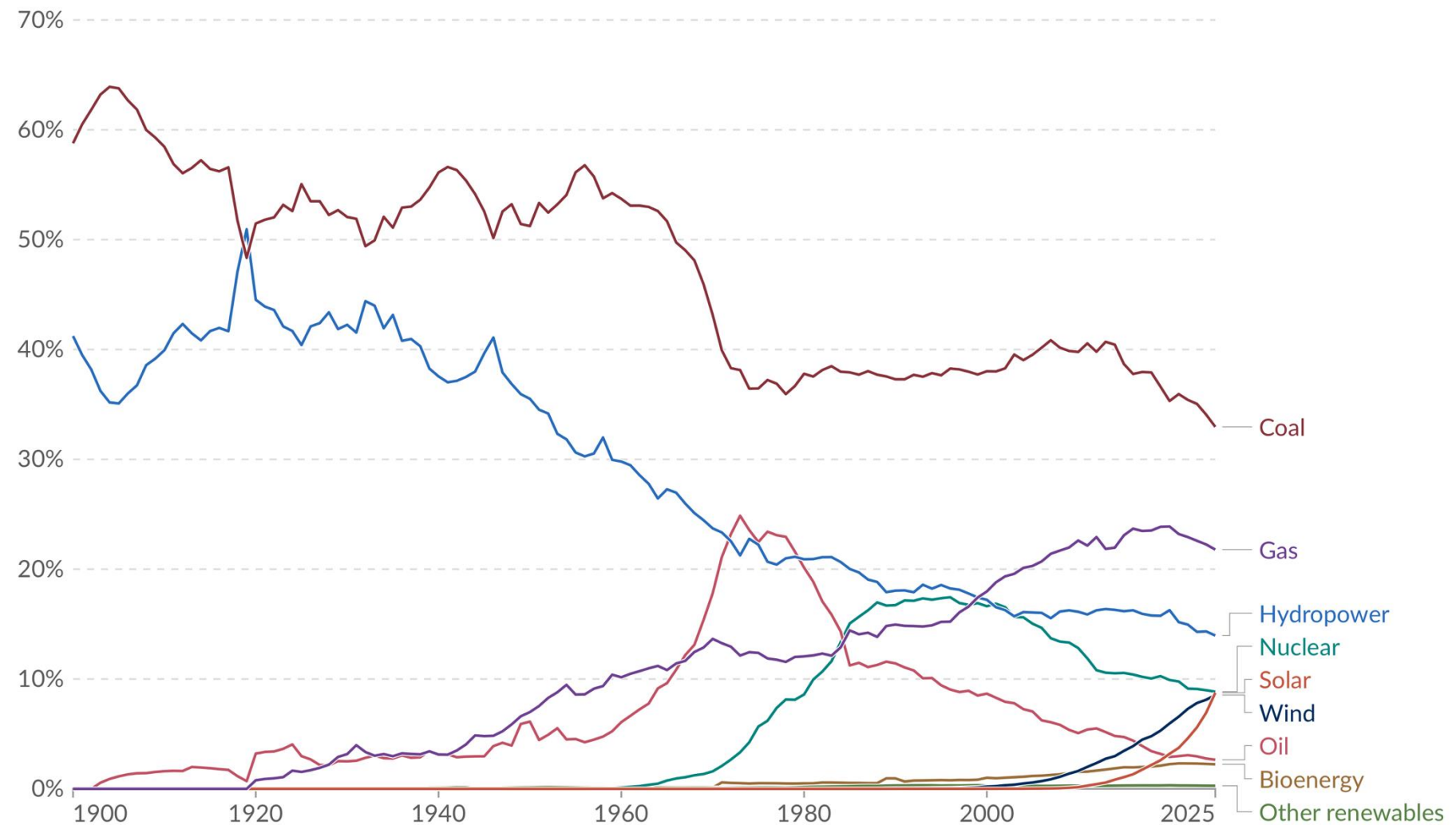
- We've talked about how to generate electricity with a turbine
- We haven't talked about what happens internally to the sources
- A few are simple:
 - Wind power : wind itself turns the turbine
 - Geothermal : steam from the earth turns the turbine
 - Fossil fuels : we burn something, it produces heat, it heats water, and steam turns the turbine
- Three merit special discussion:
 - Solar power : uses quantum mechanical properties of materials and electromagnetic radiation (i.e. light) from the sun to generate electricity directly without a turbine, called the ["photoelectric effect"](#)
 - Nuclear fission : we start from a heavy isotope, force it to radioactively decay via [nuclear fission](#), and that releases energy that can be used to heat water, steam turns the turbine
 - Nuclear fusion : we start from a light isotope, force nuclei to fuse together, and that releases energy that can be used to heat water, steam turns the turbine



The Grid Mix

- Many different sources of electricity generation, and it varies a lot by country and region
- Each have their own pros and cons, and economic viability

Share of electricity production by source World



Data source: Ember (2026) and other sources

Note: Other renewables include geothermal, wave, and tidal.

OurWorldinData.org/energy | CC BY

WIND POWER



Wind Power

- Arguably the simplest and oldest source of energy generation (if we include mechanical energy)
- Wind blows, turbine turns, electricity generated. The end.
- Options:
 - On-shore
 - Off-shore
 - Residential



<https://54energy.net>
5000W generator

The San Geronio Pass wind farm in California, United States



The Gansu Wind Farm in China is the largest wind farm in the world, with a target capacity of 20,000 MW by 2020.



Wind turbines and electrical substation of Alpha Ventus Offshore Wind Farm in the North Sea



Who Makes Wind Turbines?

- Onshore wind turbines

- [Vestas](#)
- [GE Vernova](#)
- [Goldwind](#)
- [Siemens Gamesa](#)

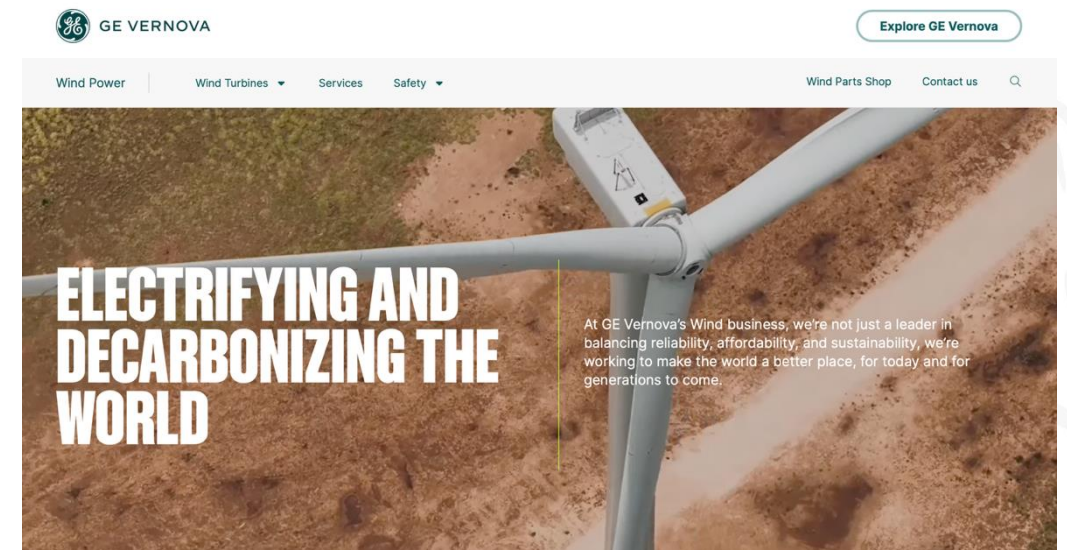
OH HAI LOOK OUR
FREN GENERAL
ELECTRIC IS BAK

- Offshore wind turbines

- [Siemens Gamesa](#)
- [GE Vernova \(Haliade-X\)](#)
- [Vestas](#)
- [MingYang Smart Energy](#)

- Residential / small wind turbines

- [Bergey Windpower](#)
- [Ryse Energy](#)
- [54energy](#)



[Home turbines from 54energy](#)

Used Claude to find
these references

What Are The Health and Environmental Risks From Wind Turbines?

- Some ecological impact from on-shore wind farms ([birds and bats, particularly](#), and another [link here](#))
- [Recycling used turbines is limited](#), but this is also present for other industrial waste
- Coatings from offshore wind turbines can [increase microplastics in the sea](#), but small compared to that from tires, for instance
- [Rare earth elements](#) are needed for the magnets, but also present for other turbines
- Installation of wind turbines has similar risks to other large-scale construction projects



<http://www.shutterstock.com/pic.mhtml?id=92928802&src=id>

<https://www.motherjones.com/environment/2014/01/birds-bats-wind-turbines-deadly-collisions/>

Used Claude to find
these references

Obvious problem with wind : sometimes it doesn't blow

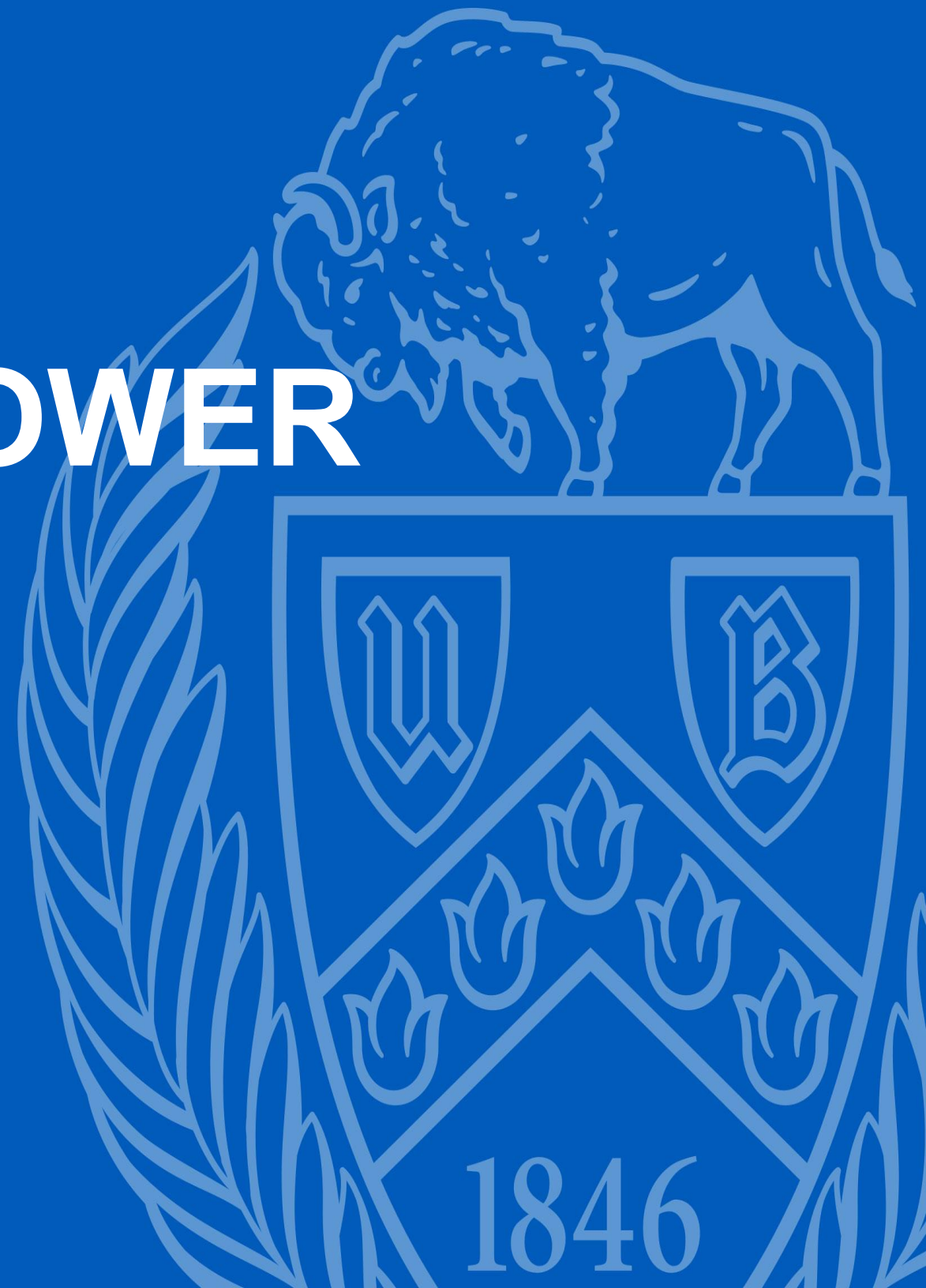
- Wind doesn't always blow so it needs supplementing
 - Batteries or use something else
- Wind is therefore intricately linked with battery usage (we'll discuss those in the future)



<http://www.shutterstock.com/pic.mhtml?id=92928802&src=id>

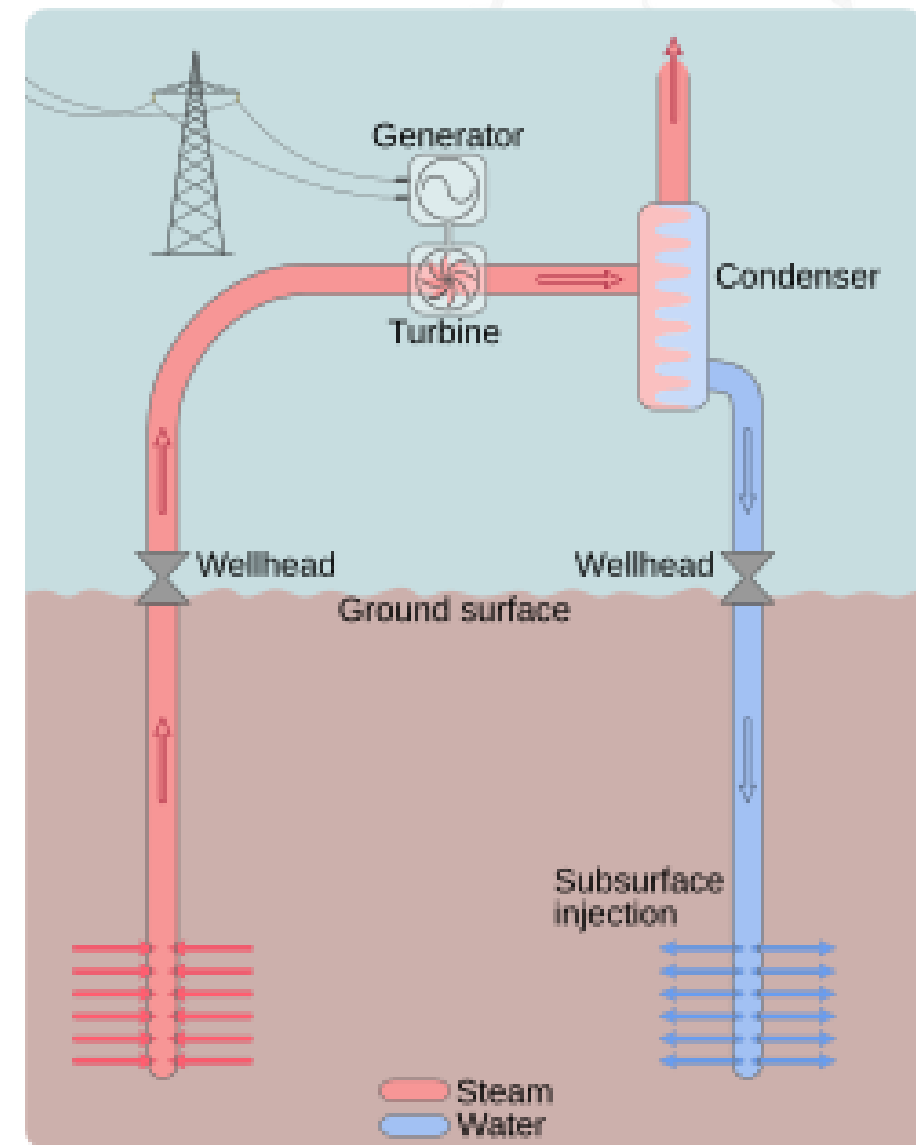
<https://www.motherjones.com/environment/2014/01/birds-bats-wind-turbines-deadly-collisions/>

GEO THERMAL POWER (AND HEAT!)



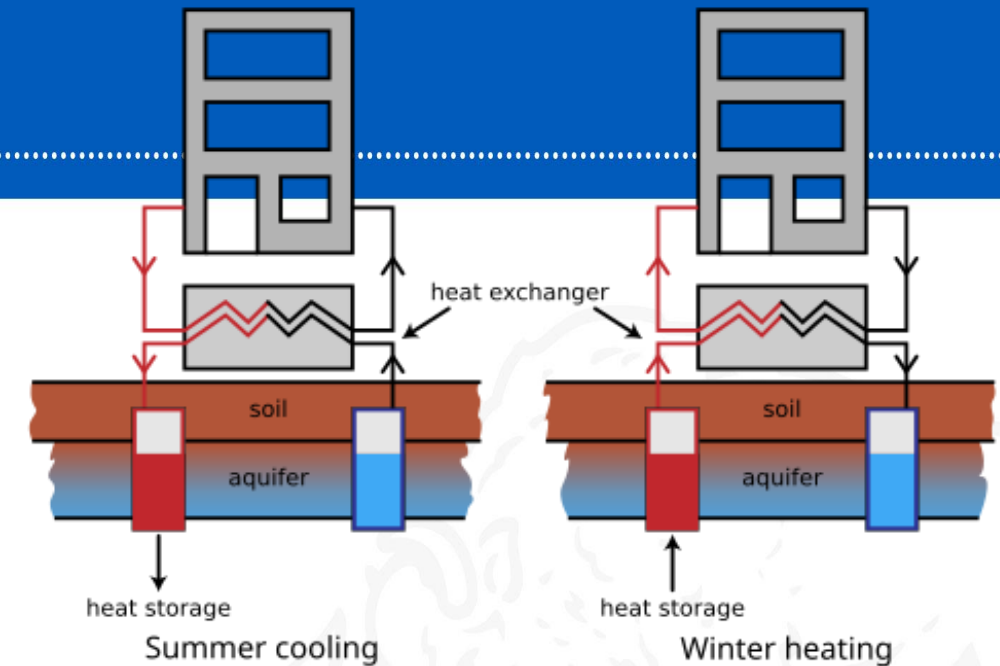
Geothermal Electricity Generation

- Geothermal power generation uses steam or heat from the earth itself to turn turbines and generate electricity
- This is a very efficient process, but has a high up-front cost to dig the wells
- Also limited to where there are steam pockets (like near geysers) or where there are pockets of magma below the surface



Geothermal Heat Pumps

- The ground can be used to heat and cool your house
 - Surface layer (1 m) : sensitive to temperature changes
 - Shallow layer (8-20 m) : slow response to temperature due to insulating higher layers
 - Deeper layer (>20 m) : effectively constant temperature
- Horizontally or vertically install pipes with a coolant to heat or cool your house depending on season



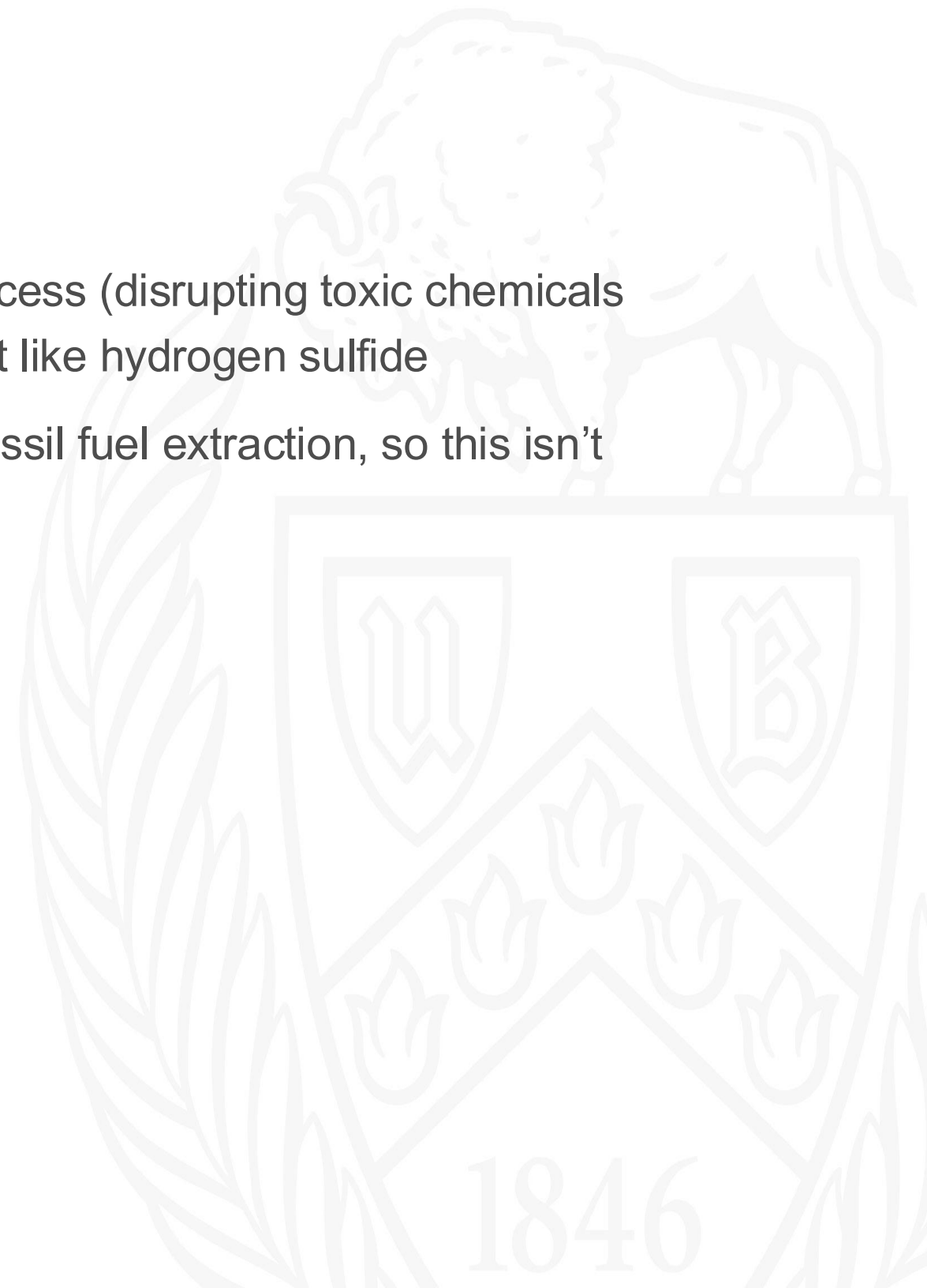
A horizontal slinky loop prior to being covered with soil



Drilling of a borehole for residential heating

Health Risks

- [Health risks of geothermal](#) are mostly in the digging process (disrupting toxic chemicals in the earth) and any outgassing of gases we don't want like hydrogen sulfide
- These are also obviously present in a larger extent in fossil fuel extraction, so this isn't unique to geothermal



Where Do Geothermal Pumps Come From?

- **Geothermal generators (power-plant turbines/generators)**

- [Ormat Technologies](#)
- [Mitsubishi Power](#)
- [Toshiba Energy Systems](#)
- [Fuji Electric](#)

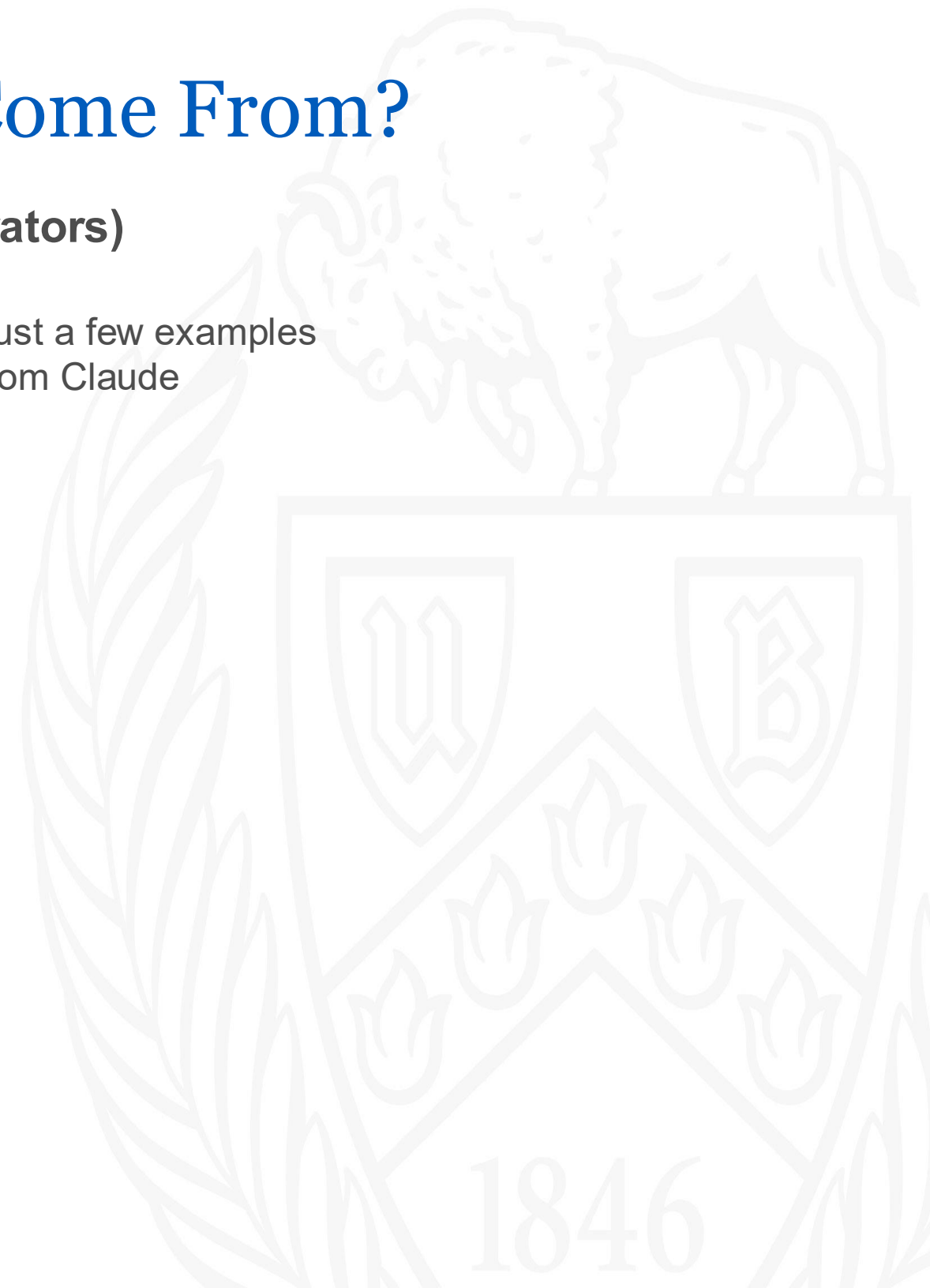
Just a few examples
from Claude

- **Geothermal (ground-source) heat pumps**

- [WaterFurnace](#)
- [ClimateMaster](#)
- [Bosch Home Comfort](#)

- **Geothermal well pumps (downhole/lifting systems)**

- [Halliburton – GeoESP](#)
- [Baker Hughes – Geothermal Solutions](#)
- [Borets – Geothermal ESP Systems](#)



FOSSIL FUELS

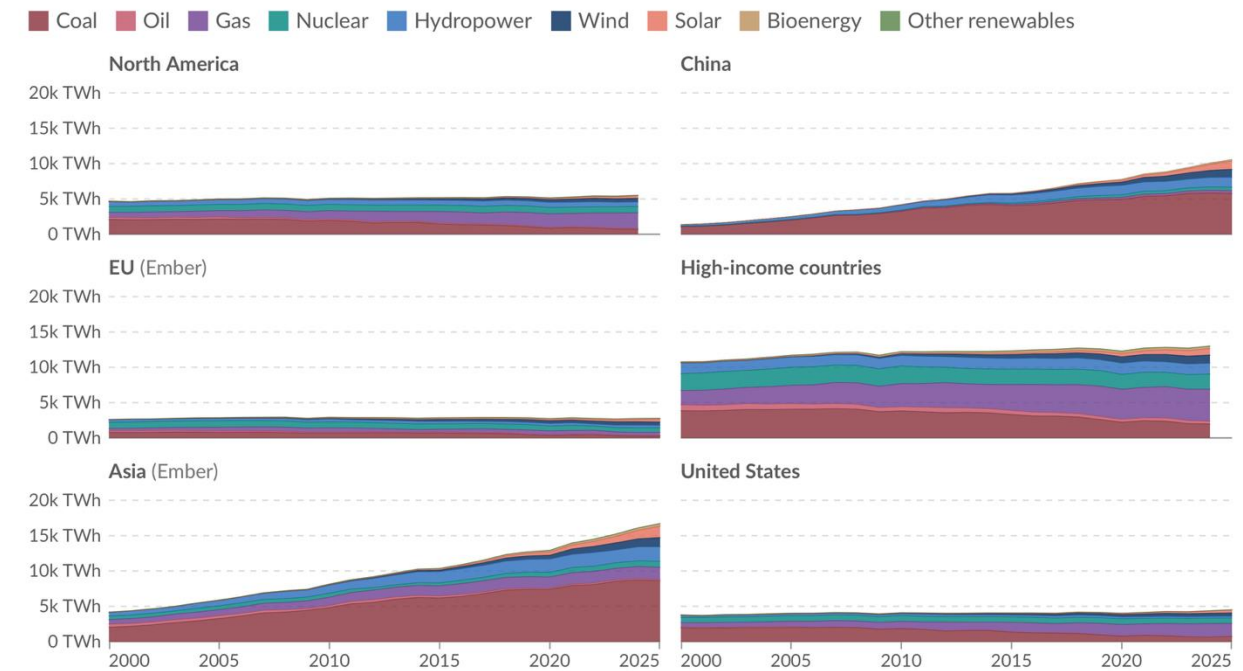


Coal Plants

- Anthracite coal has been used to power electric plants since the 19th century
- Globally, still the most popular form of power generation
- We've discussed the mining operations as well as the War of the Currents
- Coal from PA was burned to generate power across the US (and exported to other nations)
- Currently, China has surpassed the US as the leading coal consumer

Electricity generation by source 2000 to 2025

Measured in terawatt-hours¹.



Data source: Ember (2026) and other sources

OurWorldinData.org/electricity-mix | CC BY

Note: Other renewables include geothermal, wave, and tidal; bioenergy may be included prior to 2000 due to limited data availability.

1. **Watt-hour** A watt-hour is the energy one watt of power delivers for one hour. Since one watt equals one joule per second, a watt-hour equals 3600 joules of energy.

Metric prefixes are used for multiples of the unit, usually:

- kilowatt-hours (kWh), or a thousand watt-hours;
- Megawatt-hours (MWh), or a million watt-hours;
- Gigawatt-hours (GWh), or a billion watt-hours;
- Terawatt-hours (TWh), or a trillion watt-hours.

<https://ourworldindata.org/electricity-mix>

Oil and Natural Gas

- Petroleum products are one of the most widespread fossil fuels for transportation and other energy usages
 - Power plants
 - Transportation
 - Heating
 - Cooking
 - Jet fuel
- Different types of petroleum products are used for different purposes
 - Gas (petrol) : Automobiles
 - Diesel : Trucks
 - Kerosene : light
 - Propane (and Propane Accessories) : Cooking



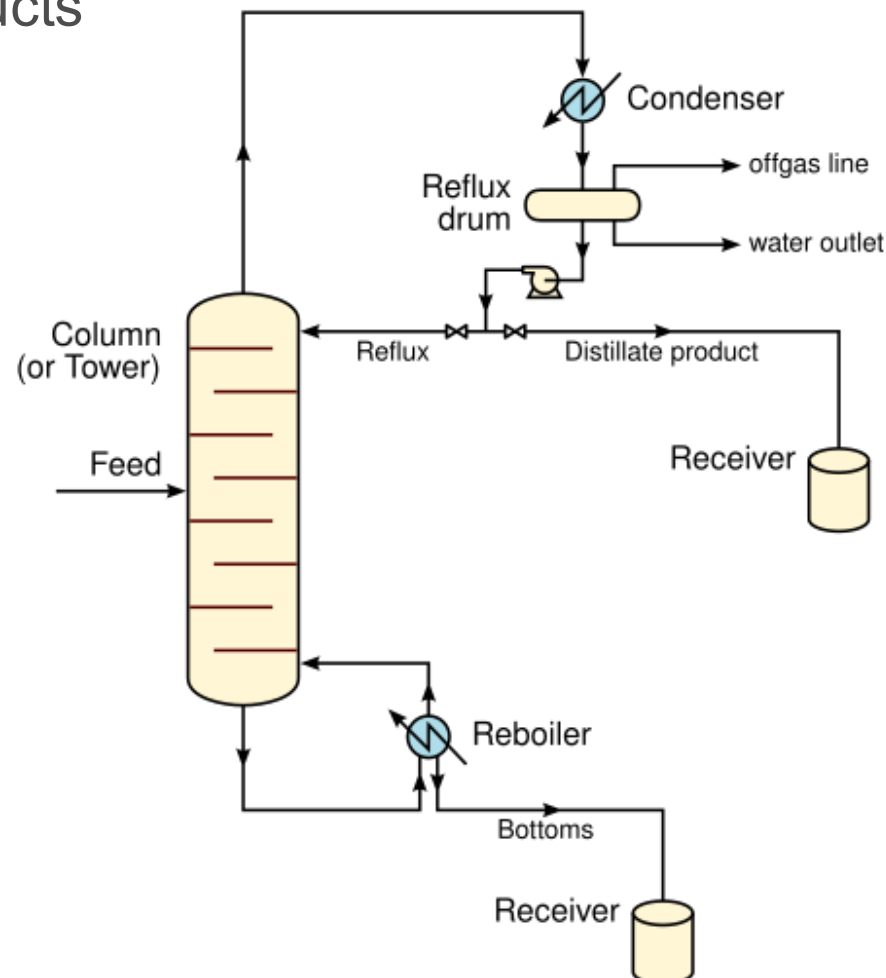
<https://www.britannica.com/science/kerosene>



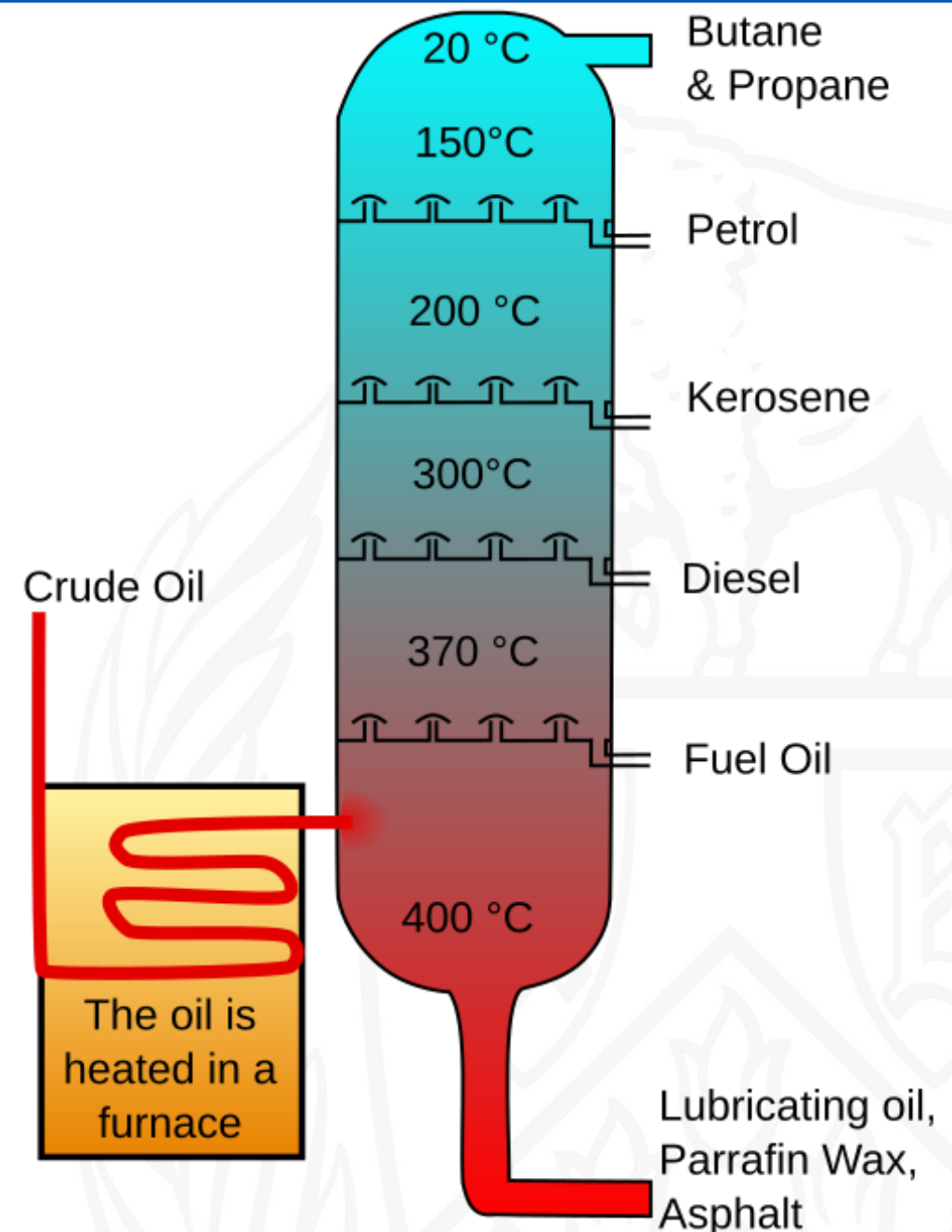
https://kingofthehill.fandom.com/wiki/Strickland_Propane

Refining

- Refining takes crude oil and turns it into usable products like gas (petrol), diesel, etc.
- Heat crude oil in a furnace, and [fractional distillation](#) is used to extract the various products



https://en.wikipedia.org/wiki/Fractional_distillation



[Crude oil](#) is separated into fractions by [fractional distillation](#). The fractions at the top of the [fractionating column](#) have lower [boiling points](#) than the fractions at the bottom. The heavy bottom fractions are often [cracked](#) into lighter, more useful products. All of the fractions are processed further in other refining units. From https://en.wikipedia.org/wiki/Oil_refinery

Gasoline and Diesel

- Gasoline is what you use to drive your car
- Diesel is what trucks use (and has more impact on the price of goods)
 - Can be used for cars, is often used in Europe



<https://www.npr.org/2026/04/07/nx-s1-5775907/high-gas-prices-save-money-iran-war>

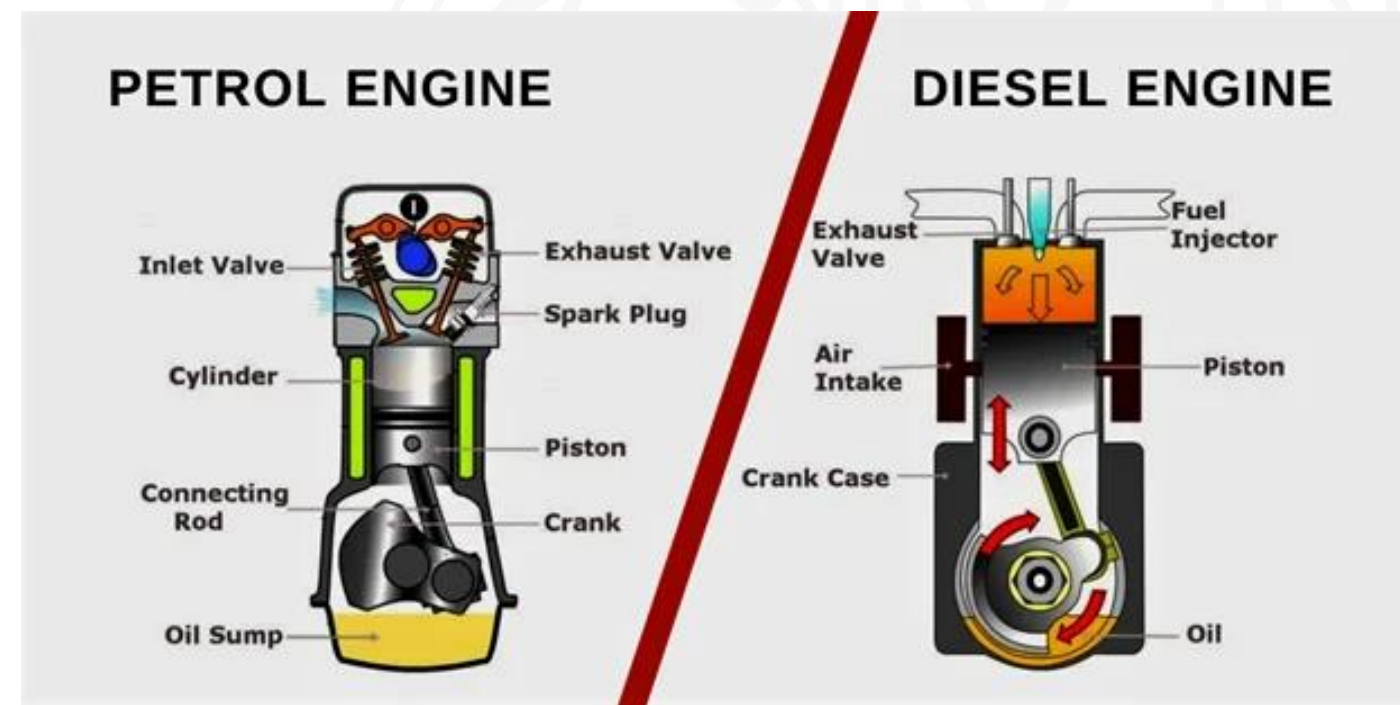


<https://www.reuters.com/commentary/reuters-open-interest/trumps-diesel-export-dilemma-good-politics-bad-economics-bousso-2026-09-23/>

Gasoline and Diesel

- Gasoline is what you use to drive your car
 - Uses a spark plug-based ignition
 - Thinner, can burn hotter, so has less efficiency but higher power output
 - Quieter engines
- Diesel is what trucks use (and has more impact on the price of goods)
 - Uses compression-based ignition
 - Thicker and more energy dense, burns at a lower energy, higher efficiency but lower power output
 - Louder engines

In gasoline engines, fuel mixes with air first and gets compressed by pistons. Then it's ignited when spark plugs create a spark. In diesel engines, air is compressed before it gets mixed with the fuel. Then fuel is injected into the hot compressed air, causing it to ignite without any spark.



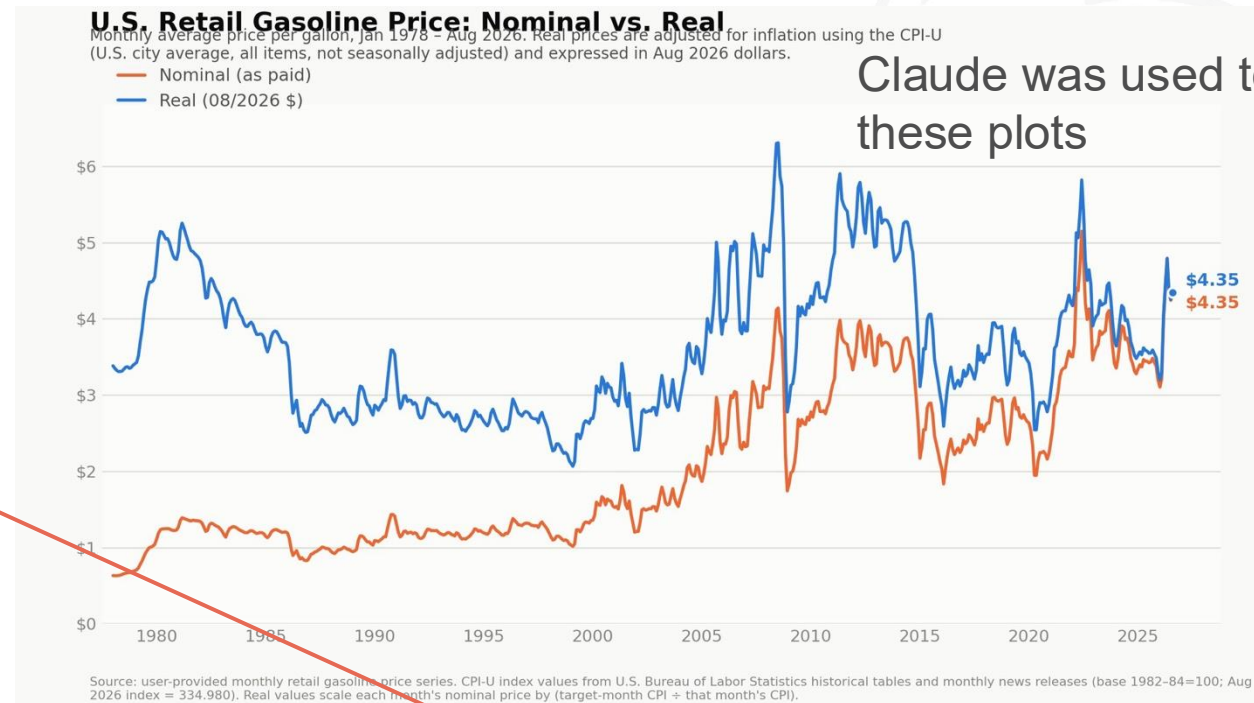
<https://www.valvolineglobal.com/en/what-is-the-difference-between-diesel-and-petrol-engines/>

Geopolitics and the Price of Gas in the US

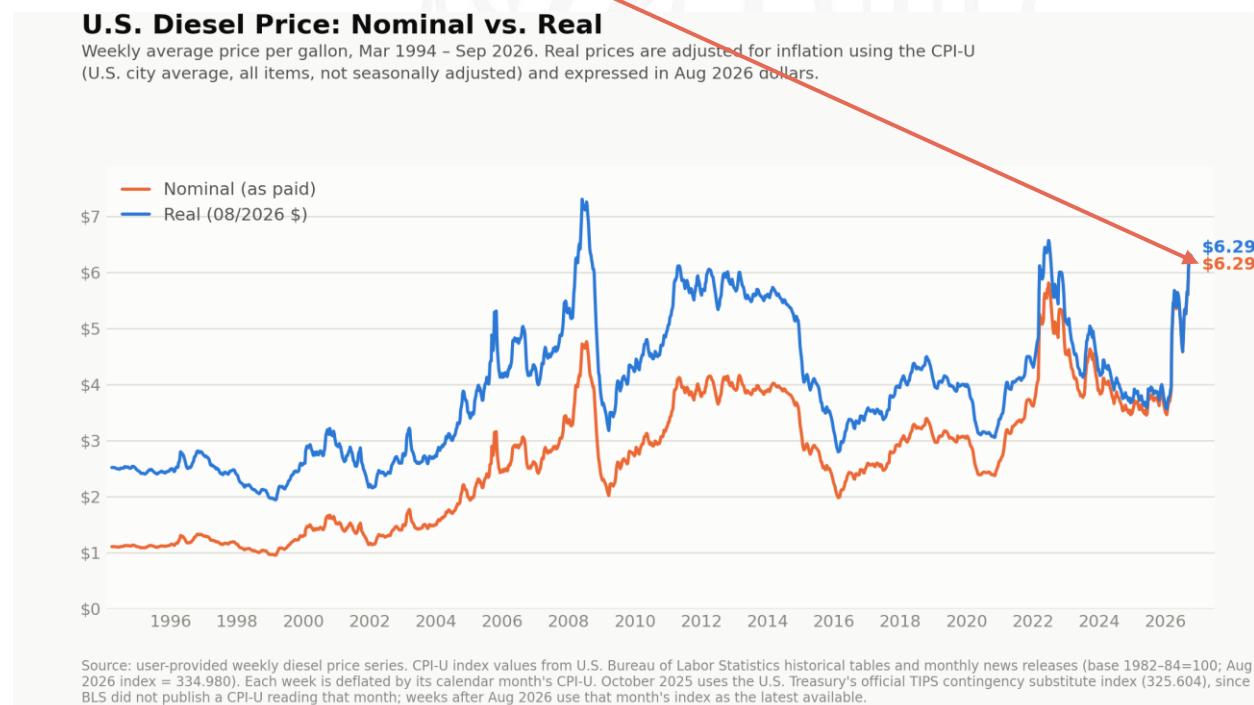
- So. Much. Geopolitics. To. Cover.
- We'll go into it later, it's a lot
- Every bump in this graph is a story
- We are currently at the highest dollar price of diesel in US history because the Strait of Hormuz is closed



https://en.wikipedia.org/wiki/2026_Strait_of_Hormuz_crisis



<https://www.macrotrends.net/4453/us-gasoline-prices>



Jet Fuel

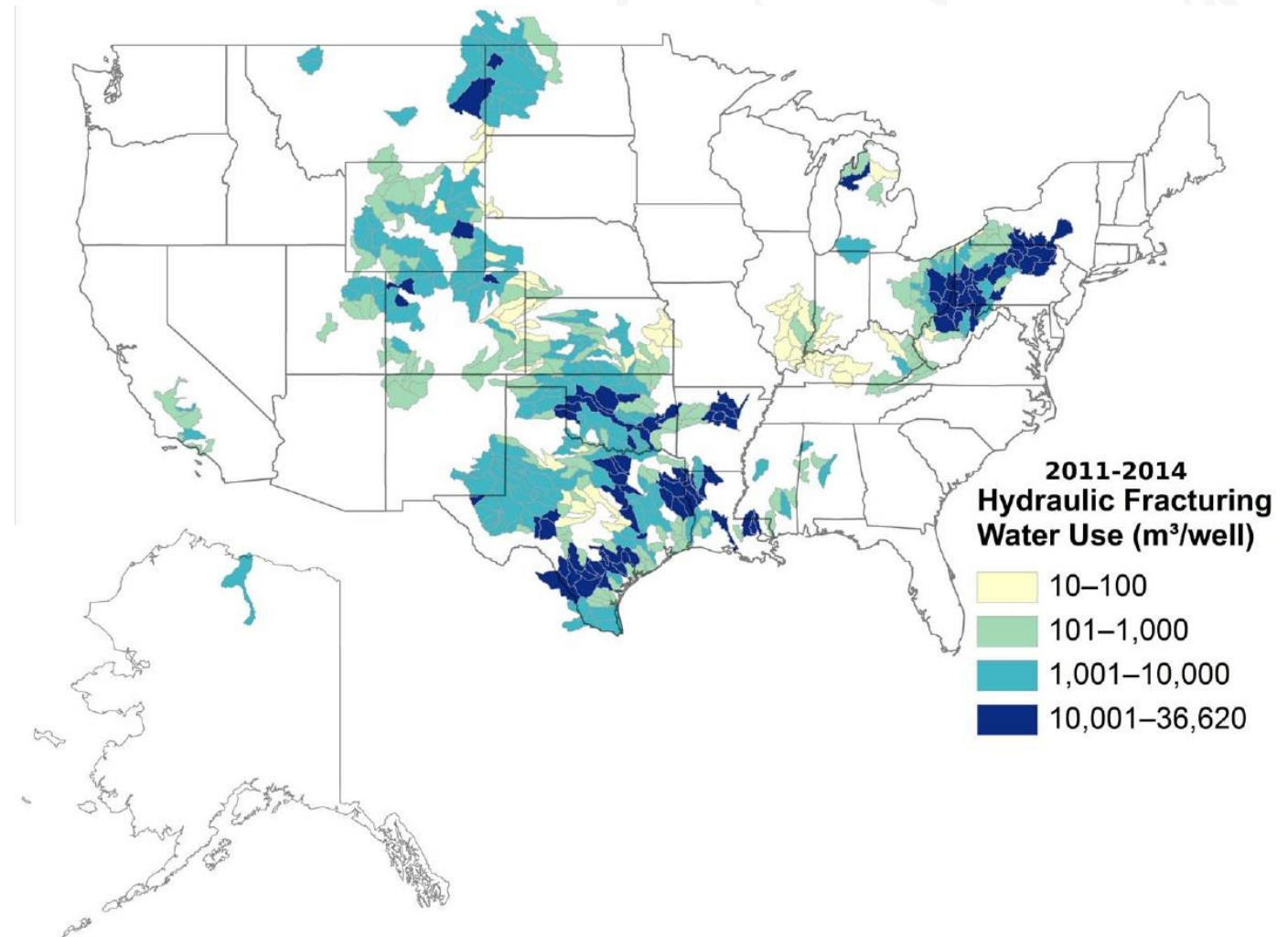
- Another critical petroleum product is jet fuel
- Air travel is one of the tougher things to solve about climate change
 - No such thing as an “electric plane”
 - Finding alternatives to air travel is ideal (trains are abundant in other countries besides the US for political reasons)
 - Other weirder options have been proposed
 - Blimps
 - Carbon offsets
- We'll discuss issues of transportation in another lecture



https://en.wikipedia.org/wiki/Boeing_747-400

Hydraulic Fracturing (Hydrofracking)

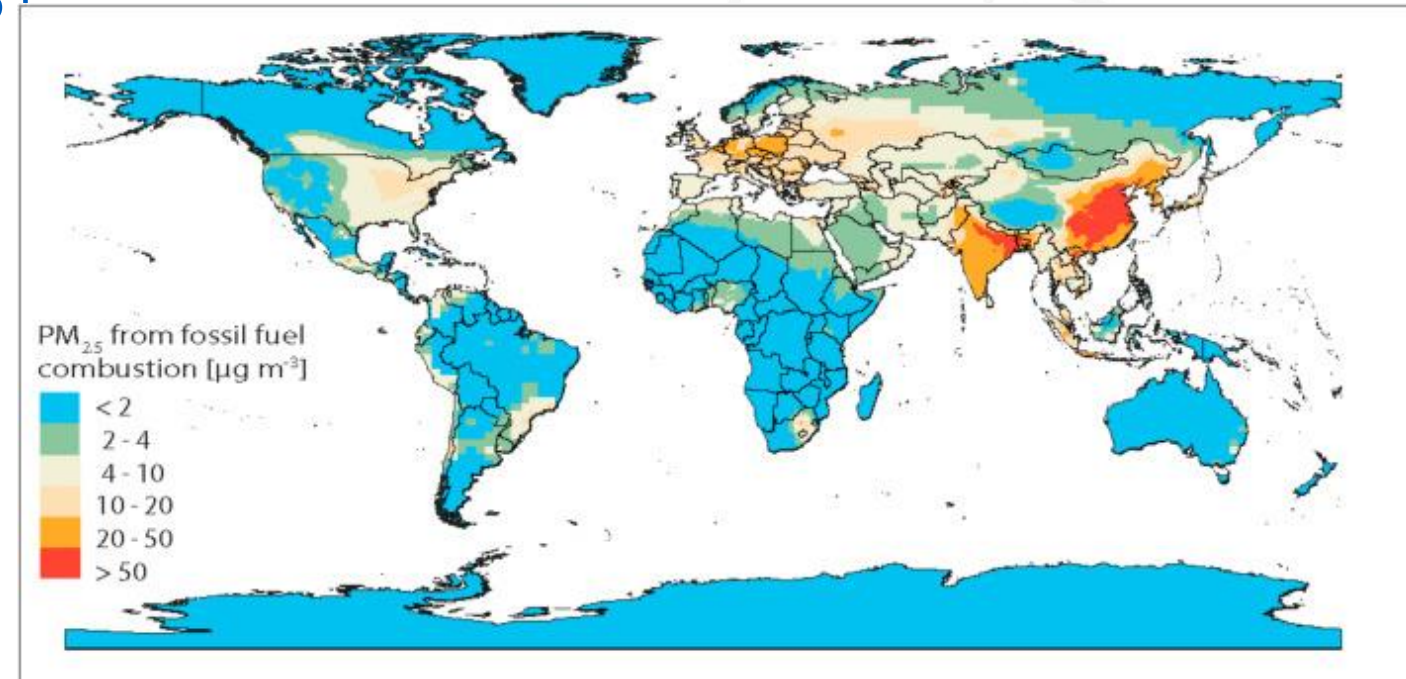
- Fracking involves pumping fluid (sandy water or acid) into bedrock to extract natural gas
- After the advent of hydraulic fracturing (hydrofracking), natural gas replaced coal as the most used fuel for power generation in the US
- Concerns include destabilizing bedrock as well as contamination of ground water



<https://en.wikipedia.org/wiki/Fracking>

What Are The Health and Environmental Risks From Burning Fossil Fuels?

- Deaths from air pollution : [8.3M deaths / year](#), [1 in 5 deaths worldwide](#)
- Impact on ecosystems : [Impact on freshwater ecosystems](#), [Impact on global fisheries and aquaculture](#), [Impact on coral reef communities via ocean acidification](#), [Global mass coral bleaching from ocean warming](#), [Amazon rainforest transformation from climate change and deforestation](#), [Pollinator \(bumblebee\) decline linked to climate change](#), [Arctic permafrost thaw and carbon cycle feedbacks](#), [Ocean deoxygenation and expanding "dead zones"](#), [Climate change–driven species extinctions](#), [Nitrogen deposition from fossil fuel combustion damaging land ecosystems](#), [Mangrove and coastal wetland loss from sea-level rise](#), [Marine heatwaves collapsing kelp forests](#), [Bird migration decoupling from spring plant growth](#), [Marine species range shifts driven by ocean warming](#), [Rising forest fire emissions linked to climate change](#)
- Carbon emissions (of course)
- Much, much more



Too many to name. We'll have a separate lecture about this.

Claude was used to gather these citations

Where Do Fossil Fuels Come From?

- This is a global, \$8T dollar (Trillion with a T) industry that is one of the most geopolitical drivers of the last 150 years. This is BIG MONEY.
- [ExxonMobil](#), [Chevron](#), [Shell](#), [BP](#), [TotalEnergies](#), [ConocoPhillips](#), [Occidental Petroleum](#), [EOG Resources](#), [Peabody Energy](#), [Glencore](#), [Coal India](#), [Saudi Aramco](#), [National Iranian Oil Company](#), [QatarEnergy](#), [Kuwait Petroleum Corporation](#), [ADNOC](#), [Rosneft](#), [Gazprom](#), [PetroChina](#), [CNOOC](#), [Sinopec](#), [Petrobras](#), [Pemex](#), [Marathon Petroleum](#), [Valero Energy](#), [Phillips 66](#), [Reliance Industries](#), [Enterprise Products Partners](#), [Kinder Morgan](#), [Williams Companies](#), [Energy Transfer](#), [Vitol](#), [Trafigura](#), [Southern Company](#), [Duke Energy](#), [American Electric Power \(AEP\)](#), [China Huaneng Group](#), [Dow](#), [BASF](#)

Used Claude to find this reference

SOLAR POWER



Solar Power and the Photoelectric Effect

- Electromagnetic radiation is, quantum mechanically, just photons traveling in space
- Photons transmit the electromagnetic force and can carry energy proportional to their frequency (f) or inversely to their wavelength (λ)

Energy:
1 Joule = 1 kg m²/s²

Planck's constant

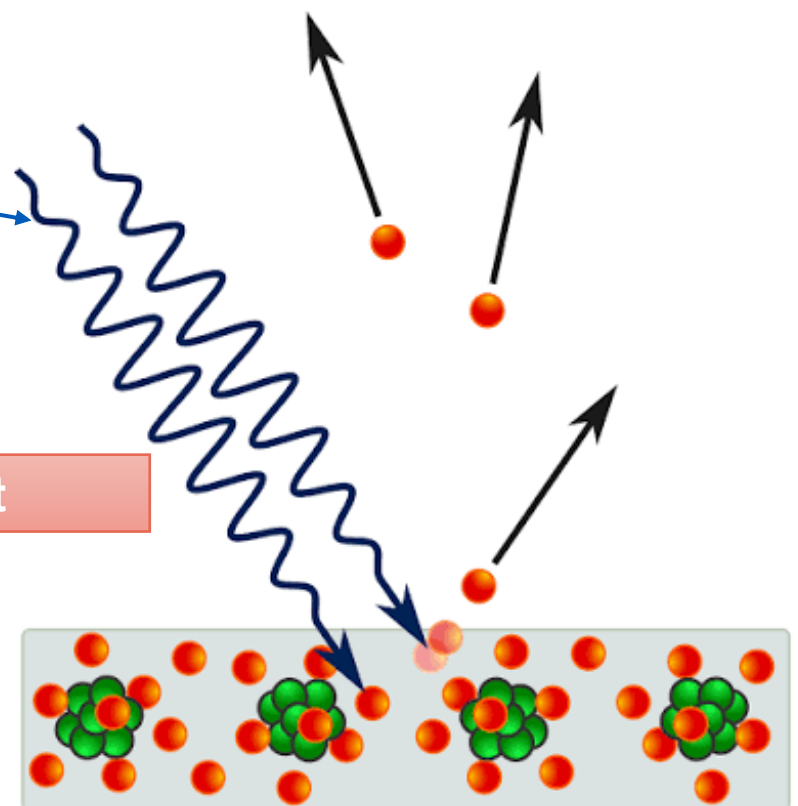
Speed of light

$$E = hf = \frac{hc}{\lambda}$$

Frequency:
1 Hertz = 1 cycle / second

Wavelength:
meters

Not responsible for this, just for enrichment



Energy in Physics

- We use Joules or kilowatt-hours as a unit of energy because we are familiar with a macroscopic, nonrelativistic world
- They're arbitrary units in physics itself, though
- We often use more natural ones where energy is in electron-volts (eV)

$$E = qV$$

$$E = \text{charge} \times \text{voltage}$$

You ARE responsible to understand that energy can be measured in electron-volts

Constants of Nature

- Planck's constant: how much energy does light have?
 - Below that special wavelength, particles act like waves

$$h = 4.135 \times 10^{-15} \text{ eV / Hz}$$

Very tiny number!
Photons don't have much energy

- Speed of light: how fast does light travel?
 - Fundamental speed limit of the universe

$$c = 299,792,458 \text{ m / s}$$

Very big number!
Photons move very quickly



Constants of nature

- Notice this formula for the energy of a photon though:

$$E = hf = \frac{hc}{\lambda}$$

- The product of the speed of light and Planck's constant occurs very frequently in physics because of this relationship, and it turns out to be pretty easy to remember:

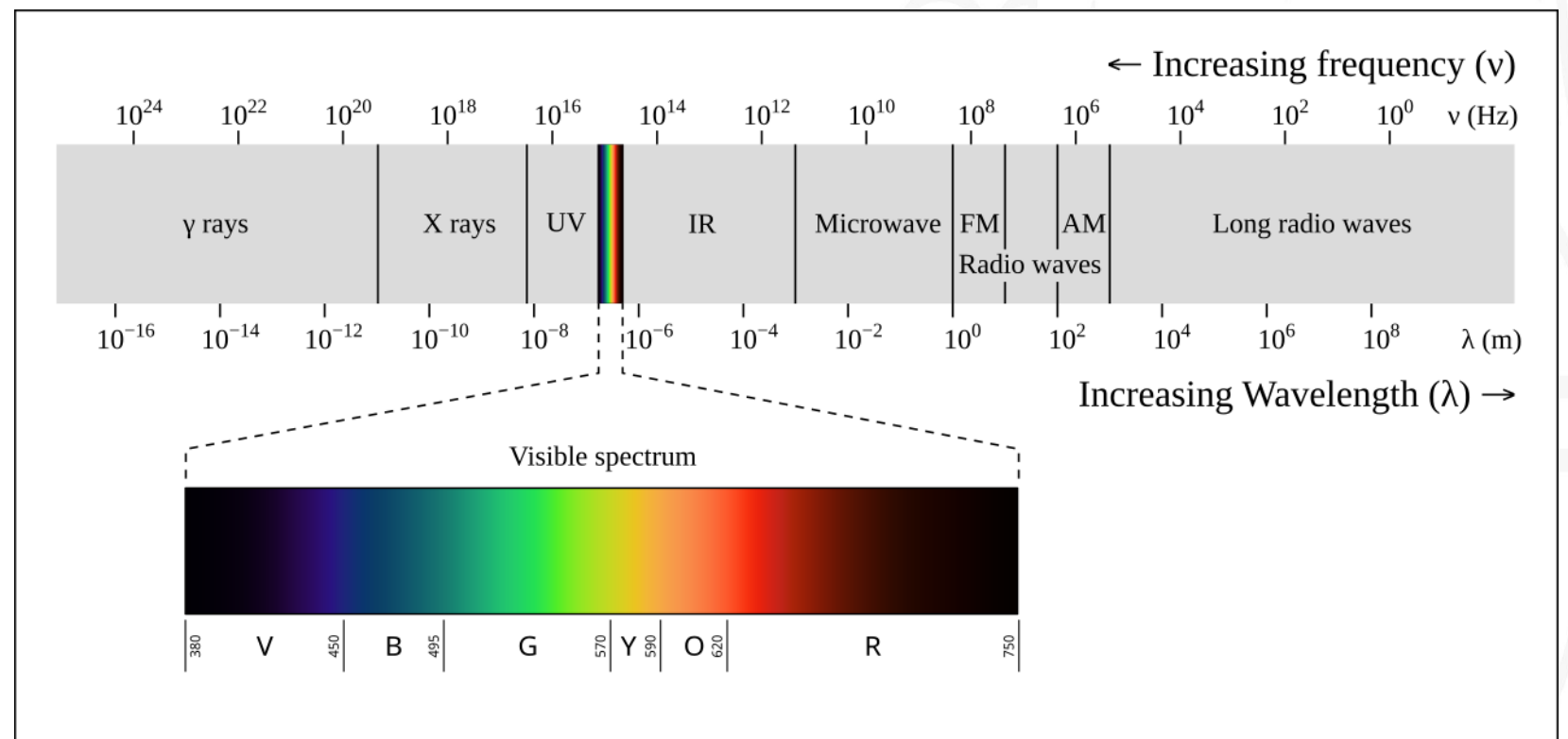
$$hc = 1240 \text{ eV} \cdot \text{nm}$$

- If you plug the photon's wavelength in nanometers, you just divide 1240 by that number, and it's the energy of the photon

You ARE responsible to understand that energy of a photon is inversely proportional to its wavelength

Electromagnetic Radiation

- We have different names for different wavelengths of light
 - Gamma (γ) rays
 - X rays
 - Ultraviolet waves
 - Visible light
 - Infrared waves
 - Microwaves
 - Radio waves
 - Long radio waves



Arrange these in order of increasing energy

- X rays
- Gamma rays
- Radio waves
- Microwaves
- Visible light



Constants of Nature

- The units we're working with are kind of inconvenient
- The trick we often use is to use units such that :
 - $c = 1$
 - We quote speeds in terms of a fraction of the speed of light, like $v = 0.3 c$
 - $\hbar = 1$
 - Define energy of a photon as exactly inversely proportional to its wavelength

Energy:
1 Electronvolt (eV)

$$E = \frac{1}{\lambda}$$

Wavelength:
Inverse electronvolts (1/eV)

- If you need to convert back to nanometers, remember $hc = 1240 \text{ eV} - \text{nm}$!

Energy-Mass Relation

- Special relativity shows that the REST energy of a massive particle is equal to the product of its mass and the speed of light squared

Mass:
Electronvolts / c^2

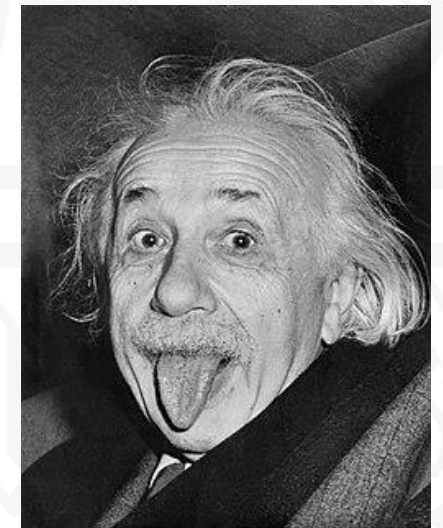
Speed of light:
 $c=1$

Energy:
1 Electronvolt (eV)

$$E = mc^2$$

Or in our better units

$$E = m$$

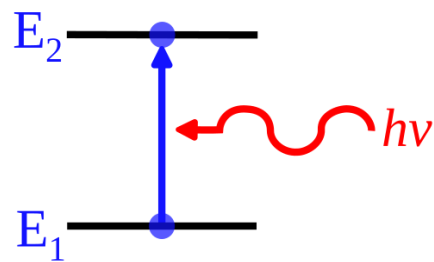


- Proved first by [Albert Einstein](#)

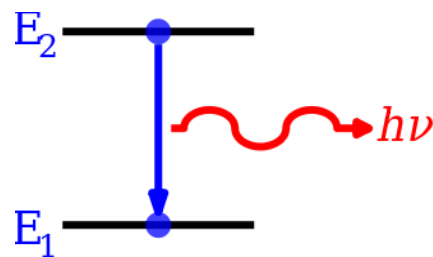
You ARE responsible to understand that energy of a massive particle is equal to mc^2

Atomic Energy Levels

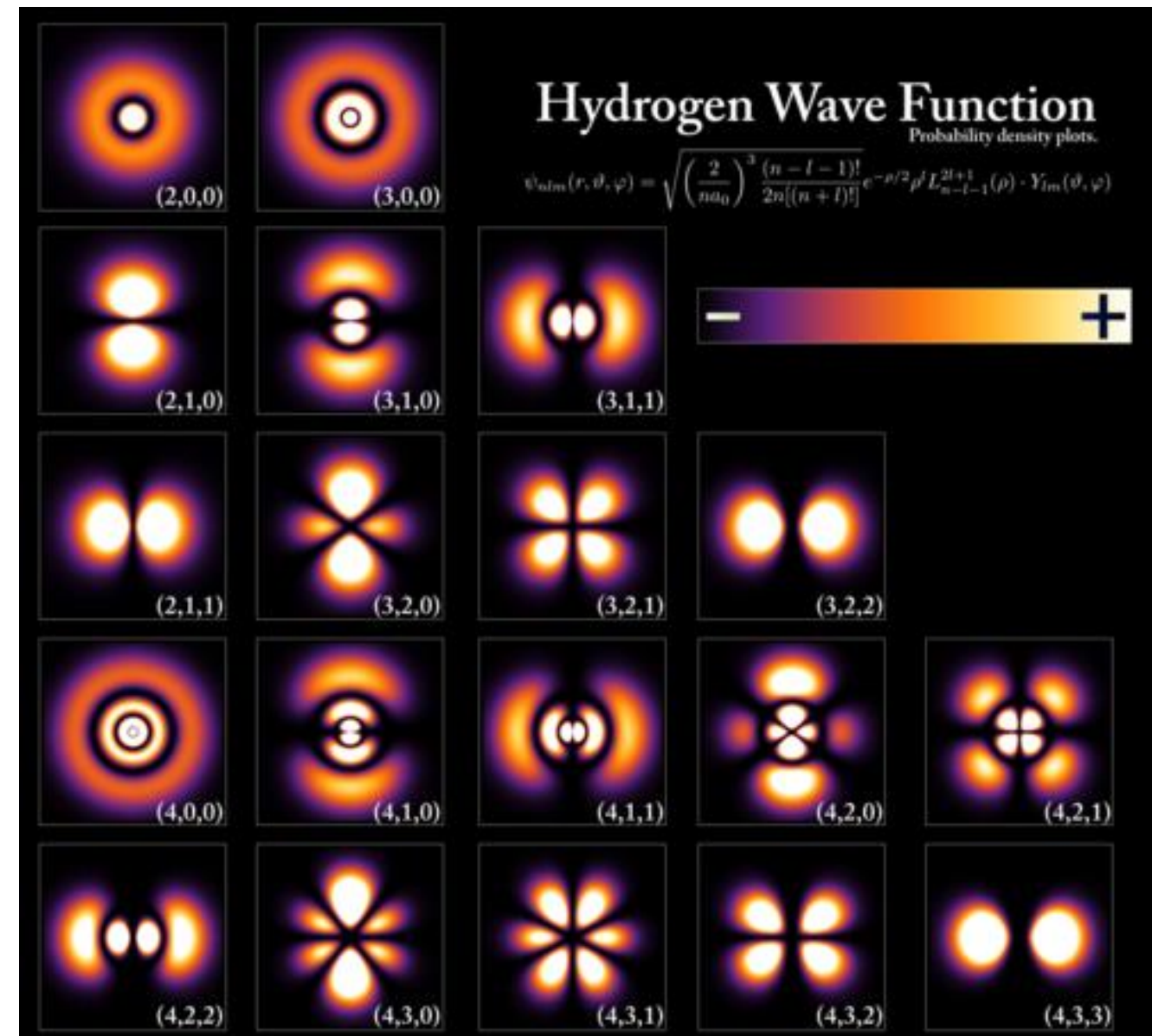
- Critically, electrons in an atomic orbital must fit a half-integer number of wavelengths in the circumference around a nucleus
- This means the energy levels are QUANTIZED, hence photons can only go up and down a specific energy by absorbing or emitting photons



An increase in energy level from E_1 to E_2 resulting from absorption of a photon represented by the red squiggly arrow, and whose energy is $h\nu$.

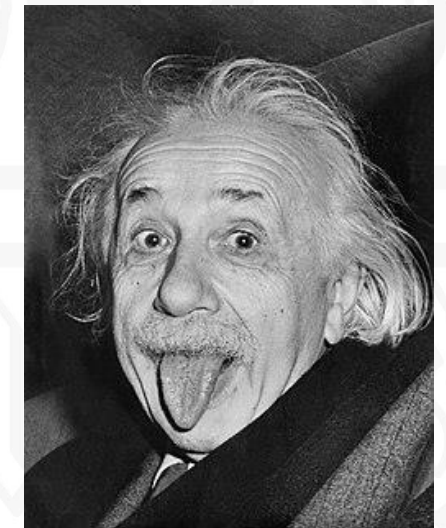
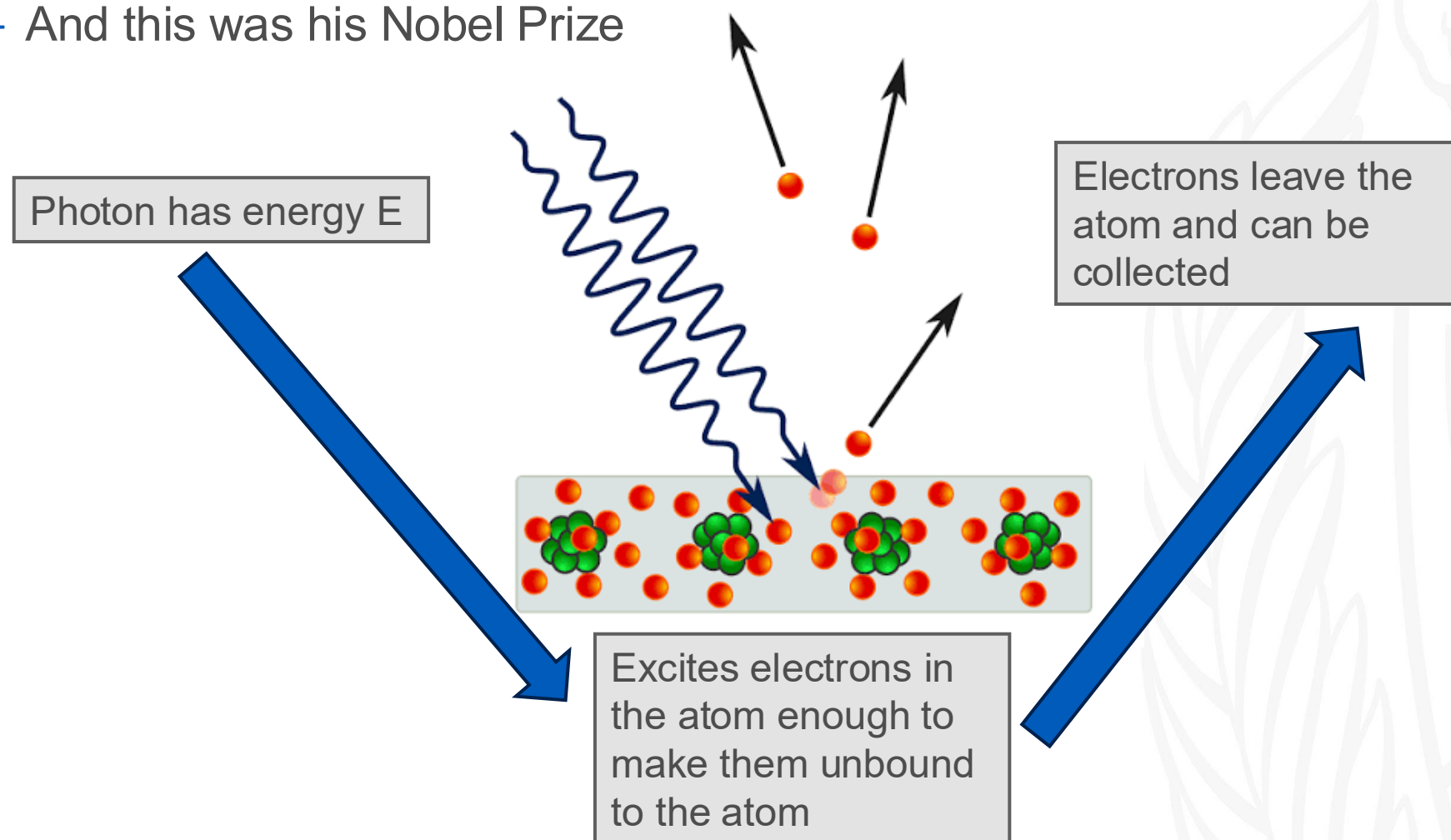


A decrease in energy level from E_2 to E_1 resulting in emission of a photon represented by the red squiggly arrow, and whose energy is $h\nu$.



Photoelectric Effect

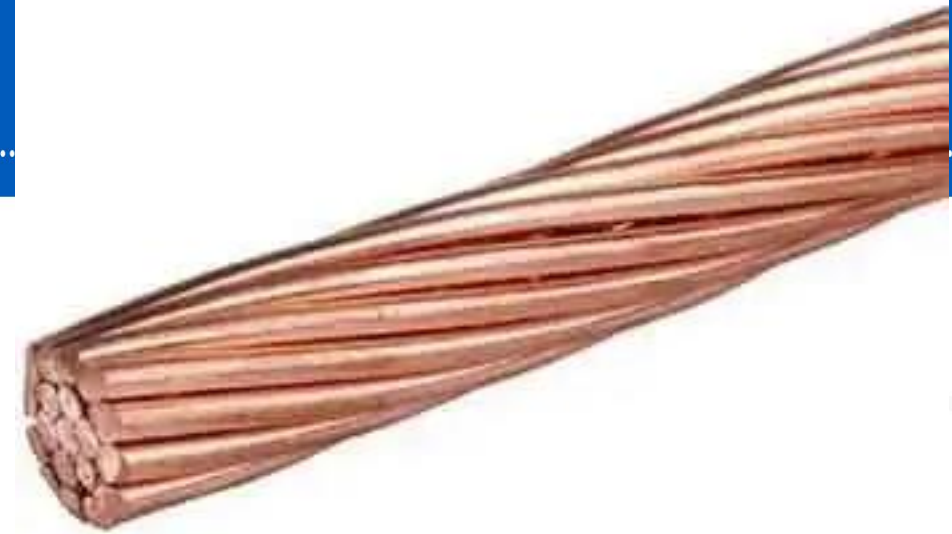
- Also proved by Einstein
 - And this was his Nobel Prize



Conductors

- In a conductor, electrons are able to move freely around the material
- As we've discussed before, they don't go very far, but the highest-energy electrons are easily able to move to nearby atoms and get captured
- This means the electrons are kind of "free agents" that don't belong to any one atom
- Most metals are conductors, with copper and gold having very high conductivities

You ARE responsible to understand that conductors allow electrons to flow



<https://electricityforum.com/what-is-a-conductor>

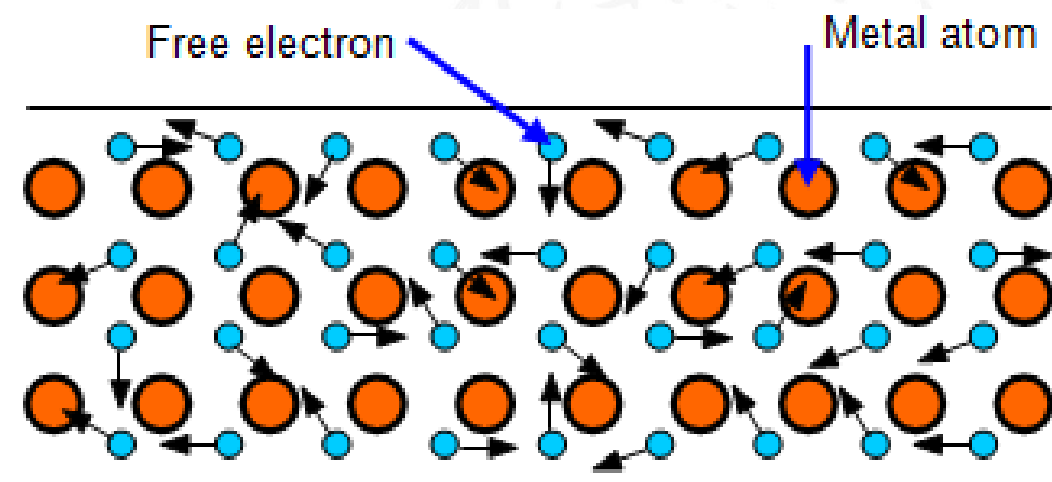


Figure 1(a)

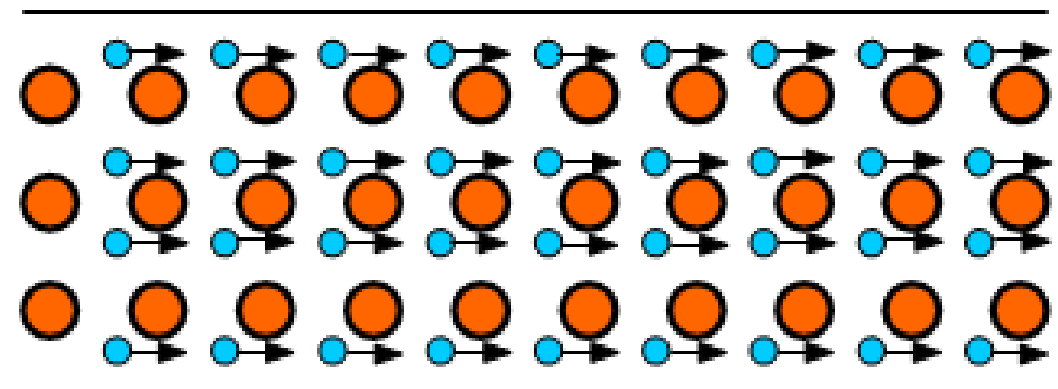
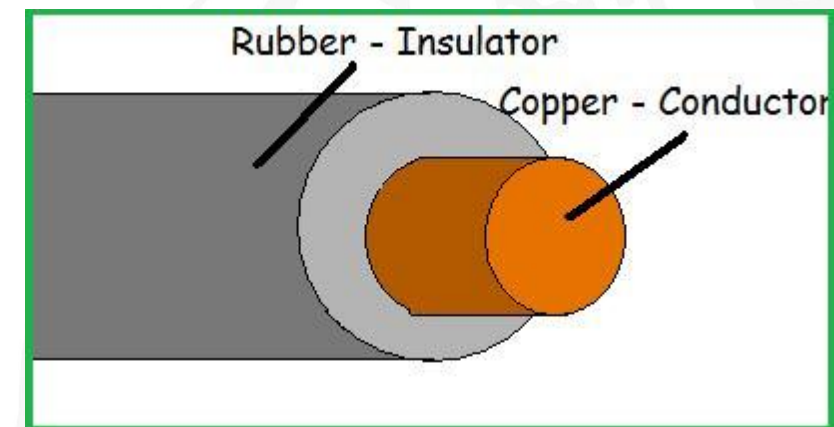


Figure 1(b)

https://www.schoolphysics.co.uk/age11-14/Electricity%20and%20magnetism/Current%20electricity/text/Electric_current/index.html

Insulators

- In an insulator, electrons are more tightly bound to their atomic cores
- They cannot move freely, so heat and electricity don't travel down the material well
- Examples are Silicon Dioxide (SiO_2 , or sand) and derivatives of it (glass, ceramics, rocks like quartz, marble, etc), most plastics (combinations of carbon and hydrogen), wood, rubber, and air itself.



<https://gusmedina.weebly.com/conductors--insulators.html>

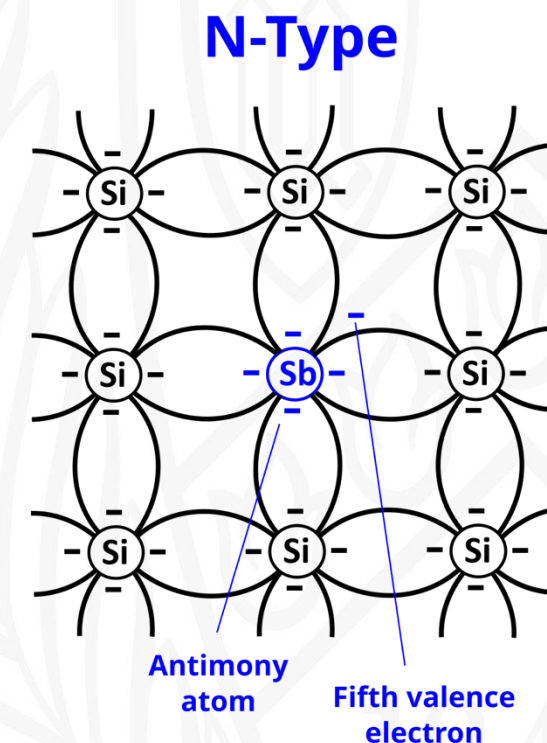
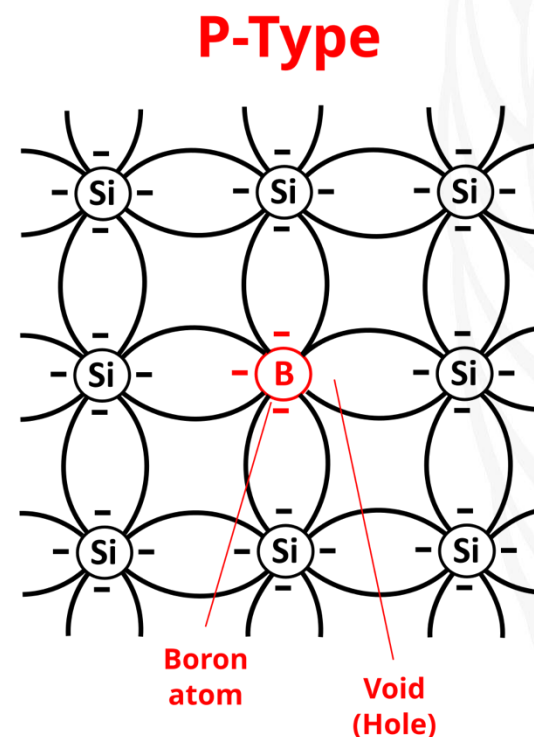
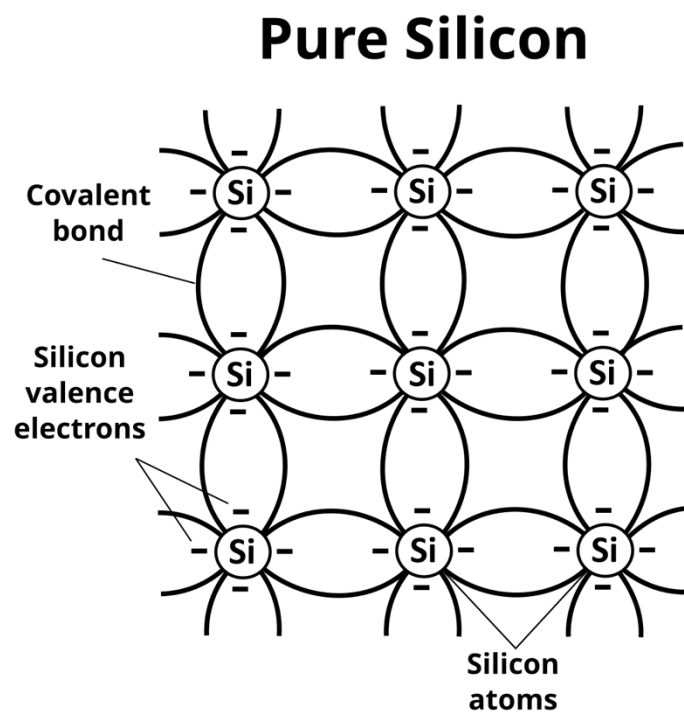
You ARE responsible to understand that insulators do not allow electrons to flow

Semiconductors

- These are insulators that can BECOME conductive under the right circumstances
- Usually a crystal lattice is DISRUPTED by adding impurities (known as “[doping](#)”)
- These add or subtract electrons that are “expected” by the lattice
- Semiconductors then join two regions with different levels of doping to create a [junction](#)

Intrinsic (Undoped)

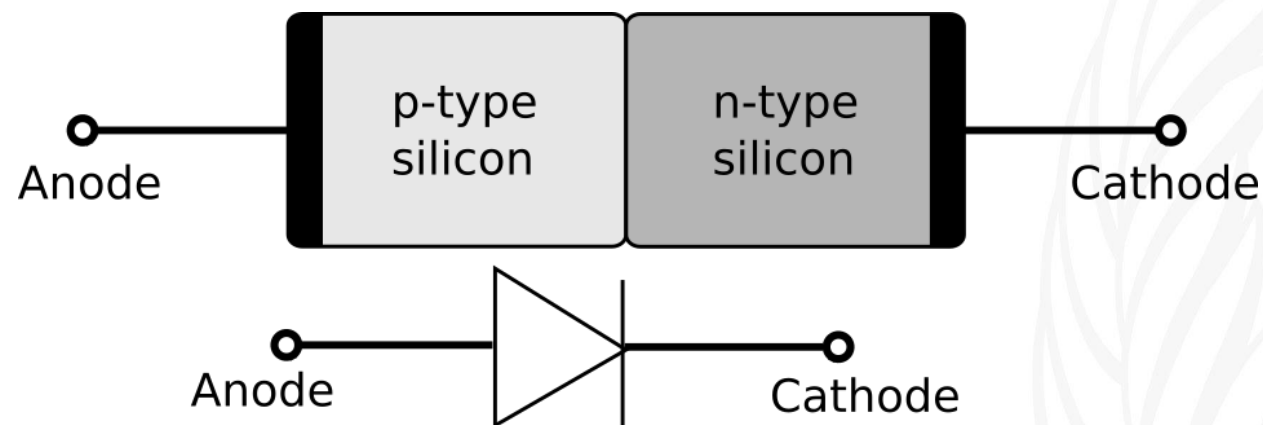
Extrinsic (Doped)



Not responsible for this, just for enrichment

Semiconductors

- If you JOIN two types of semiconductors (one that has extra electrons – n-type, one that has fewer electrons – p-type), you can apply a voltage to give the electrons a bit of an extra “kick” to get them to become conducting in ONE direction only : a [DIODE](#)



- You’ve heard of diodes, you’re looking at them right now... [light emitting diodes \(LEDs\)](#)
 - Very energy efficient because it only takes a tiny amount of energy to make the electrons move, and for an LED, it emits photons of very specific wavelengths (colors!)

Not responsible for this, just for enrichment, but

You ARE responsible to understand that diodes allow electricity to flow one way and that LEDs are light-emitting diodes.

Photovoltaic Cells

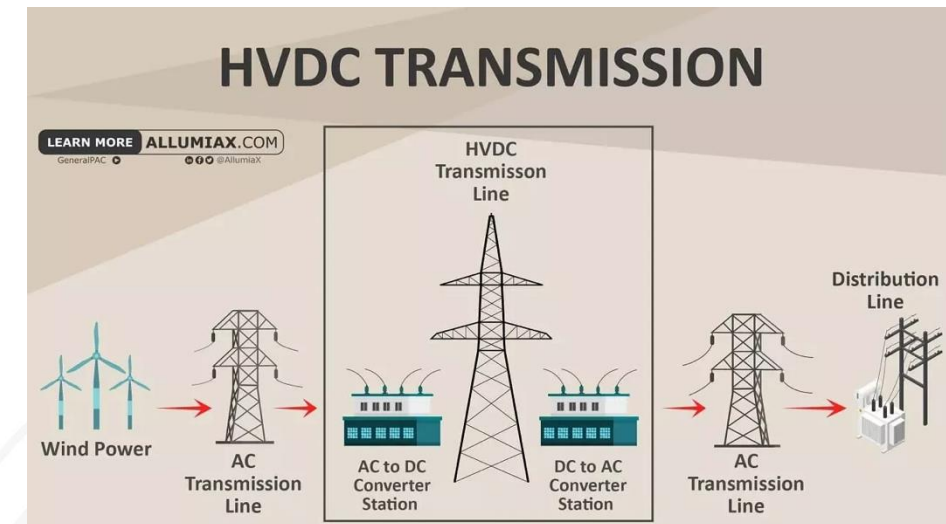
- Most are made of crystalline silicon
- **Solar cells are basically REVERSE LEDs**
- Instead of releasing photons when the voltage is applied, it absorbs them
- The energy that is captured from the source (like the sun) can be much higher than the energy used to supply the voltage, so you get a direct source of energy
- **But it's DIRECT CURRENT (DC)**



Left side: solar cells made of polycrystalline silicon Right side: polysilicon rod (top) and chunks (bottom)

DC and the Grid

- To use DC power generators, we would need to either:
 - Change the power from DC to AC
 - Incurs a little bit of loss
 - Change the grid from AC to DC
 - Which we already know is generally a bad idea from the War of the Currents
 - However high-voltage DC power lines (HVDC) can be used in specific circumstances
 - These avoid our Ohmic (i^2R) losses by increasing the voltage considerably above AC systems
 - For VERY long distances, HVDC performs better than AC because the wire itself carries a capacitance and inductance that can create an AC loop WITHIN the wire
 - It requires a massive change to every building in the grid, however



<https://www.allumiax.com/blog/high-voltage-direct-current-hvdc-transmission>

For charging batteries, can use solar power directly and efficiently!

Obvious problem with solar: Nighttime.

- SOLar panels rely on... well... Sol.
- What do you do at night?
 - Batteries or use something else
- Solar panels are therefore intricately linked with battery usage (we'll discuss those in the future)
- Good news is that they are the absolute best at charging batteries!



[Sol, god of the sun in Roman mythology](#)



[Sòl, god of the sun in Norse mythology](#)

What Are the Health and Environmental Risks From Solar Panels?

- Installing rooftop solar is still a bit more dangerous than you'd expect
- Falling off roofs is a hazard.
 - Around 5000 people die each year falling off roofs
 - [2020 report](#) showed a 15% increase in deaths from installing rooftop solar

Where Do the Solar Panels Come From?

- Silicon is very abundant, but the expertise to turn it into solar panels is not
- Here are the top 5 producers, with 9 Chinese companies producing >90% of the panels in the market
 - Tongwei Group
 - GCL Technology Holdings
 - Daqo New Energy
 - Xinte Energy
 - Wacker Chemie AG

Used Claude to find this reference

To remember

Concept	Formula
Energy of a charged particle in a voltage field is charge times voltage	$E = qV$
Energy of a photon is inversely proportionate to its wavelength	$E = \frac{hc}{\lambda}$
Energy of a massive particle is equal to mc^2	$E = mc^2$
Conductors allow electrons to flow	
insulators do not allow electrons to flow	
diodes allow electricity to flow one way and that LEDs are light-emitting diodes.	

NEXT LECTURE : NUCLEAR POWER, CAPACITY FACTOR, COST

